STALKS OF PULLBACK MODULES AT PAIRED POINTS: (F^*\MATHCAL{F})_P = \MATHCAL{F}_Q \OTIMES_{\MATHCAL{O}_{\MATHCAL{D},Q}} \MATHCAL{O}_{\MATHCAL{C},P}

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Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file sites-modules.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted span cites one same-chapter lemma and two Sites lemmas, recorded in assembly.external_references with canonical tags as navigation pointers.

Allowed standard prerequisites (editorial).

- Stalks commute with tensor products. Stacks Project, Modules on Sites, Lemma lemma-tensor-product-pullback. Supplies that applying p^{-1} to a tensor product of modules gives the tensor product of the stalks.
- Stalks of pullbacks under a morphism of sites/topoi. Stacks Project, Sites, Lemmas sites-lemma-point-morphism-sites and sites-lemma-point-morphism-topoi. Supplies the identification of the stalk of f^{-1}\mathcal{F} at p with the stalk of \mathcal{F} at q.

Source reference. Original lemma 03EL: sources/stacks-chow/tags.txt, source lines 4460–4460. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 05V1: sources/stacks-chow/tags.txt, source lines 7535–7535. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 05V2: sources/stacks-chow/tags.txt, source lines 7536–7536. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let $(f, f^\sharp) : (Sh(\mathcal{C}), \mathcal{O}_\mathcal{C}) \to (Sh(\mathcal{D}), \mathcal{O}_\mathcal{D})$ be a morphism of ringed topoi or ringed sites. Let p be a point of $\mathcal{C}$ or $Sh(\mathcal{C})$ and set $q = f \circ p$. Then*

$$(f^*\mathcal{F})_p = \mathcal{F}_q \otimes_{\mathcal{O}_{\mathcal{D},q}} \mathcal{O}_{\mathcal{C},p}$$

for any $\mathcal{O}_\mathcal{D}$-module $\mathcal{F}$.

Proof. We have

$$f^*\mathcal{F} = f^{-1}\mathcal{F} \otimes_{f^{-1}\mathcal{O}_\mathcal{D}} \mathcal{O}_\mathcal{C}$$

by definition. Since taking stalks at p (i.e., applying p^{-1}) commutes with $\otimes$ by Lemma 03EL we win by the relation between the stalk of pullbacks at p and stalks at q explained in Sites, Lemma 05V1 or Sites, Lemma 05V2. □

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